



Heat Exchanger Network Design and MTBE Production

October 12th

Senior Design, Mini Project 2

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Table of Contents	Page#
1. Project Charter – Part 1	...2
2. Design Plan – Part 1	...3
3. Introduction	...4
4. PFD Before Integration	...4
5. Composite Curves	...5
6. Optimization of ΔT_{min}	...8
7. Pinch Temperature Calculation	...9
8. Heat Exchanger Network Design	...9
9. PFD After Integration	...11
10. Utility Cost	...13
11. Project Charter – Part 2	...14
12. Design Plan – Part 2	...15
13. Introduction	...16
14. Unit Balances	...16
15. Aspen Simulation with Shortcut	...19
16. Aspen Simulation with Rigorous	...21
17. Parameter Variation Results	...25
18. Azeotrope Search	...31
19. Block Flow Diagram	...32
20. Process Flow Diagram	...33
References	...34

1. Project Charter – Part 1

Title	Goal	In Scope	Out of Scope
Utility Optimization	To design a heat exchanger network by optimizing utility consumption	- Calculate optimized dTmin - Pinch temperature analysis - Heat exchanger network design - Utility cost before/after heat integration	- Calculate actual dTmin - Components/physical properties inside the streams
Deliverables		Team Members	
- Design Plan - PFD - Composite curves - Optimized dTmin - Pinch temperature calculation - Utility cost		Huda Usman Kellen Krupa Milos Markovic Joel Finlon	

2. Design Plan – Part 1

Task ID	Task	Start Date	Finish Date	Member assigned
1	PFD Before/After Integration	10/5	10/11	Kellen
2	Composite Curves	10/7	10/9	Kellen & Milos
3	Optimization of T _{min}	10/9	10/9	Huda
4	Pinch Temp	10/7	10/9	Joel
5	Heat Exchange Network Design	10/6	10/11	Joel & Huda
6	Before Integration Cost	10/10	10/11	Milos
7	After Integration Cost	10/10	10/11	Huda
8	Describe Process	10/10	10/11	Milos & Joel
9	Introduction	10/11	10/12	Joel
10	Final Editing	10/12	10/12	All

3. Introduction - Pinch, Heat, And Cool, The Optimal Cost Heat Capacity Tool

We propose the optimization of the following process using a heat exchanger network by heat integration. Heat exchangers transfer thermal energy from one fluid to another through a passive mass of metal without the fluids mixing with one another. [1] The process consists of two streams at 20 °C that are mixed and heated to 160 °C before being fed to a reactor where an endothermic reaction occurs. The product exits the reactor at 120 °C before being heated to 260 °C before distillation. The liquid product from the column must be cooled to 20 °C from 180 °C before storage while the bottom product is cooled from 280 °C to 60 °C before arriving at storage. Adopting a heat exchanger network allows for a significant reduction in required equipment and as a result reduces utility yearly operating costs and servicing.

4. Process Flow Diagram (PFD) – Before Integration

In the process presented in Figure 1, two heaters and two coolers heat the streams to target temperatures of 160°C, 260°C, 60°C, and 20°C. Two feed streams at 20°C are mixed and heated by utility 1 to 160°C before being fed to a reactor. Inside the reactor an endothermic reaction occurs, heating the product to 120°C. This stream is heated by utility 2 to 260°C and the following stream is sent to a distillation column where the product is recovered. At the top of the column, the distillate is cooled from 180°C to 20°C. The bottom product is cooled from 280°C to 60°C [2]. All stream temperatures, flow rates, specific heat capacities, and duties are summarized in Table 1.

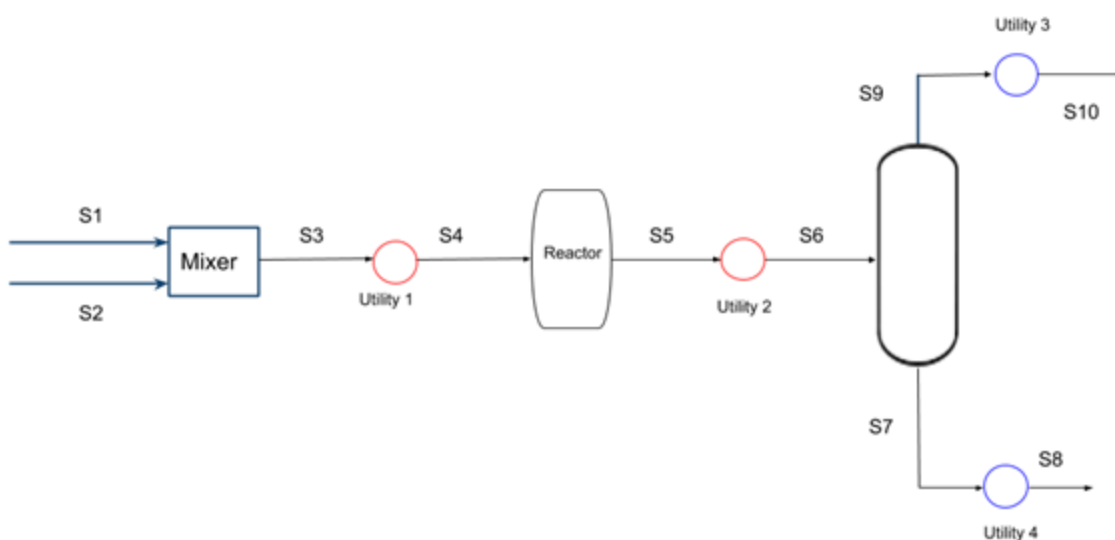


Figure 1: Flowsheet before Heat Integration

Table #1: Stream Summary Before Integration

Stream	T (°C)	Flow (kg/s)	C _p (kJ/kg °C)	Block	Δ H(kW)	Utility Type
1	20	5	5	-	-	-
2	20	5	5	-	-	-
3	20	10	5	Utility 1	7000	MP steam
4	160	10	5			
5	120	10	5.5	Utility 2	77000	MP steam
6	260	10	5.5			
7	280	4	5	Utility 4	6600	Cooling Water
8	60	4	5			
9	180	6	10	Utility 3	6400	Chilled Water
10	20	6	10			

5. Composite Curves

Composite curves were created using an Excel spreadsheet. Hot and cold stream curves were plotted separately and then combined on a composite curve graph, which was then shifted to account for the $\Delta T_{\min}$ of 20°C. Hot stream and cold stream curves are plotted in Figure 2 and Figure 3. To combine the hot stream curves together and the cold stream curves together, the change in enthalpies were calculated. These are plotted according to their corresponding temperatures. The hot composite curve is plotted over the T_{in} range from 20°C to 280°C, and the cold composite curve is plotted over the T_{out} range from 20°C to 260°C (Table 2, Table 3).

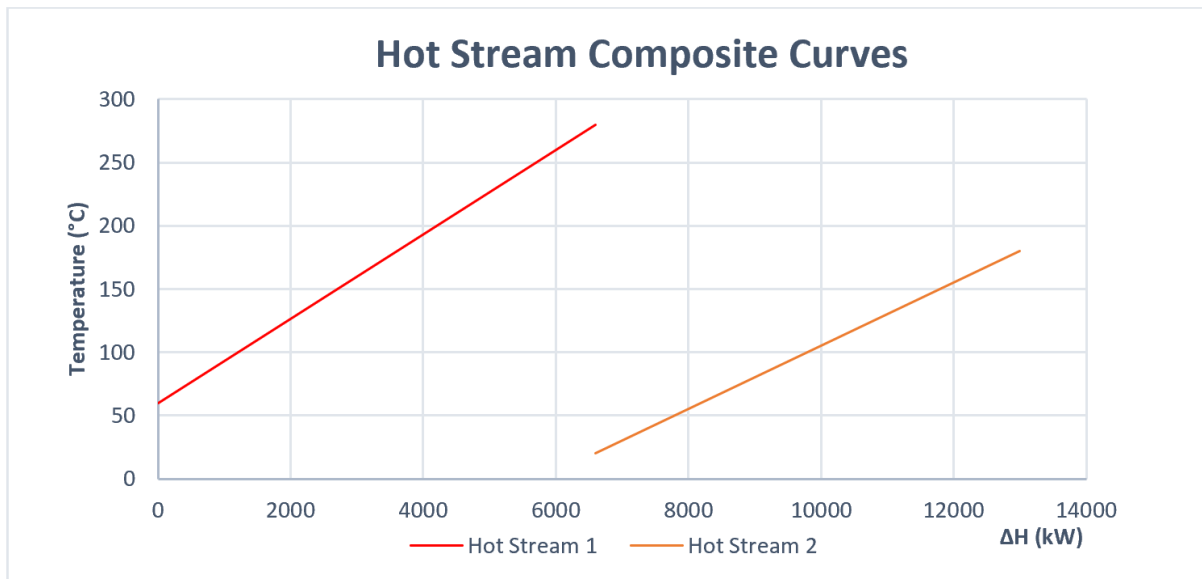


Figure 2: Hot Stream Composite Curves

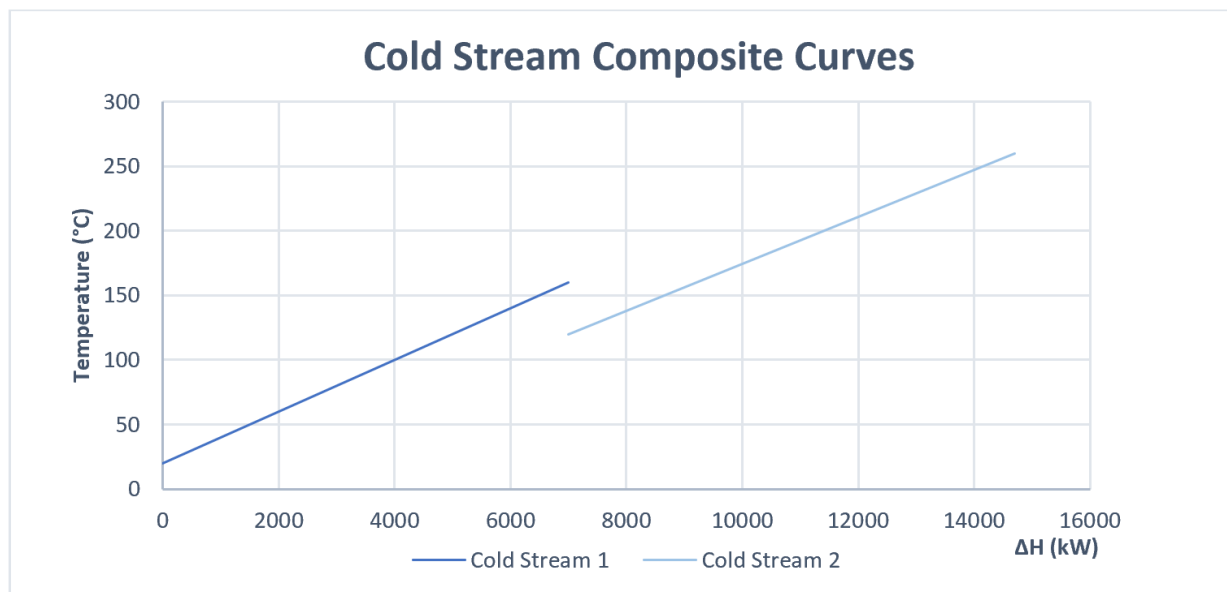


Figure 3: Cold Stream Composite Curves

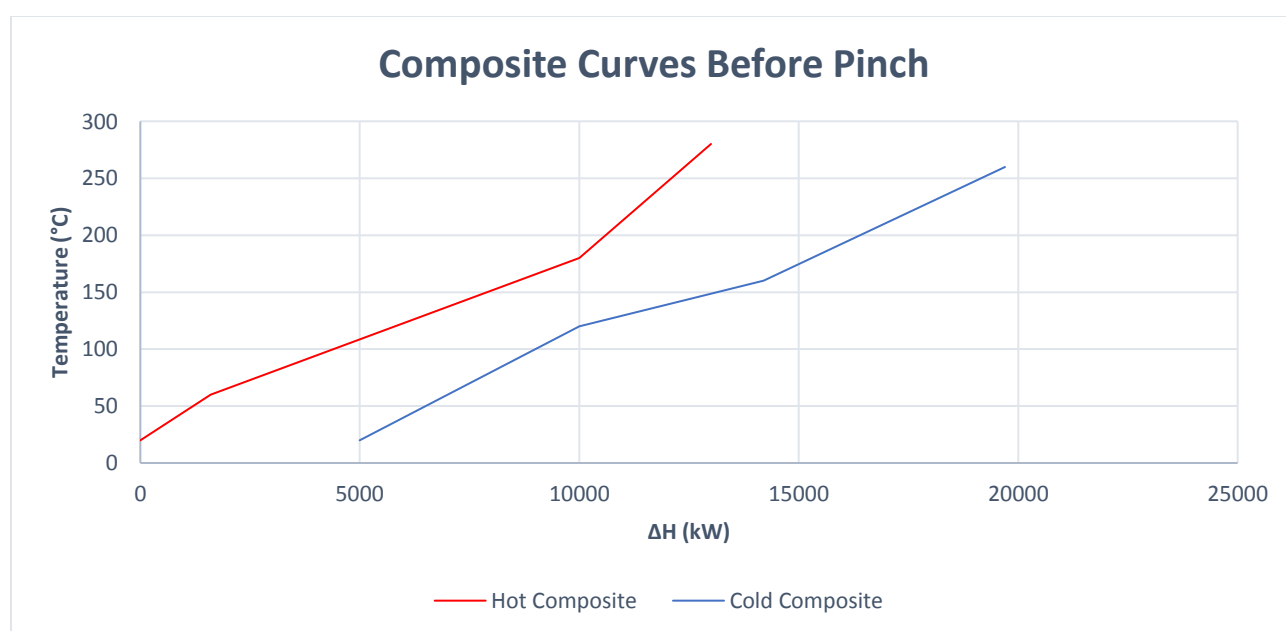
Table #2: Excel calculations for plotting hot composite stream.

Hot Composite					
Stream	T _{in}	T _{out}	dT	m*C _p	dH
HOT 1	280	180	-100	30	-13000
HOT1+HOT2	180	60	-120	70	-10000
HOT 2	60	20	-40	40	-1600

Table #3: Excel calculations for plotting cold composite stream.

Cold Composite					
Stream	T _{in}	T _{out}	dT	m*C _p	dH
COLD 2	160	260	100	55	14700
COLD1+COLD2	120	160	40	105	9200
COLD1	20	120	100	50	5000

Figure 4 displays the composite curves before the graph was shifted to create the $\Delta T_{\min}$ separation. For clarity of the graph, the cold composite curve begins at 5000 kW. True change in enthalpy values are available in Table 3.

Figure 4: Hot and cold stream composite curves before $\Delta T_{\min}$.

Since the cold composite curve initially contains a plot at 120°C and a pinch between 120°C and 140°C is desired, the hot curve is shifted to the right. To create the pinch and get the curves to line up on the same line on the x-axis, the shift value was calculated. Using the linear function of the hot composite curve found by Excel, a temperature value of 140°C and ΔH of 10000 was input and a shift value of 2807.2 was determined. This value was added to all the previous ΔH values to create new plots for the hot composite curve shown in Figure 5. The $\Delta T_{\min}$ is now satisfied from 120°C to 140°C.

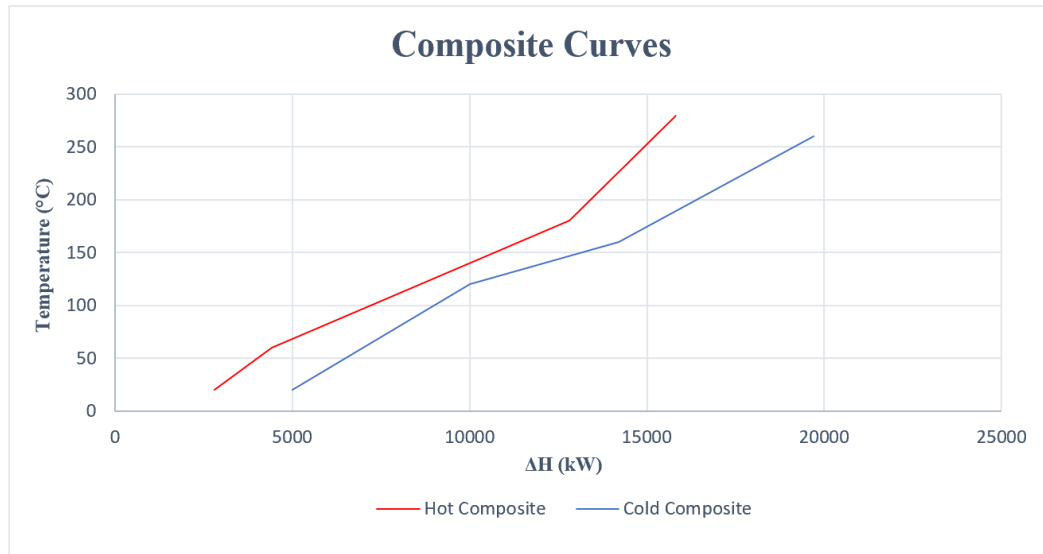


Figure 5: Hot and Cold Stream composite curves with $\Delta T_{\min}$.

6. Optimization of $\Delta T_{\min}$

To optimize $\Delta T_{\min}$, a plot of the total annual cost versus a range of $\Delta T_{\min}$ is constructed, as exhibited by Figure 6. For each $\Delta T_{\min}$, the yearly capital and operating costs are determined and summed to provide an annualized total cost. From the plot, the minimum $\Delta T_{\min}$ with the lowest total cost is selected [1]. Figure 6 shows an optimized $\Delta T_{\min}$ of 30 at the lowest total cost. In some cases, optimizing the minimum temperature will depend on the area of the heat exchanger because the cost increases with the increase in size of the exchanger. Also plotting the area targets vs $\Delta T_{\min}$ or first correlating cost versus area and then plotting a cost versus $\Delta T_{\min}$ would provide an optimized minimum temperature [3].

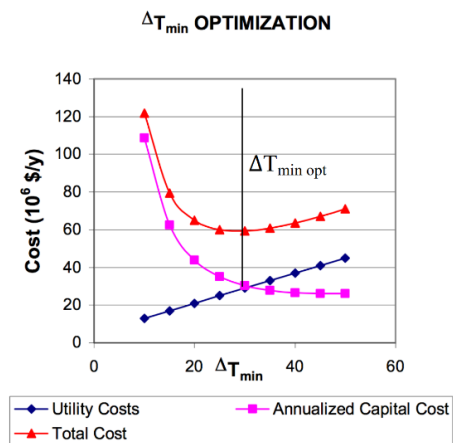


Figure 6: Example plot of annual cost vs. $\Delta T_{\min}$. [3]

7. Pinch Temperature Calculation

Pinch temperature analysis was completed using the Excel spreadsheet from Towler and Sinnott [4]. The $\Delta T_{\min}$ was provided at an optimized temperature of 20°C and all source and target temperatures, heat capacity flow rates, and heat loads were inputted into the spreadsheet. From Figure 7, under the problem table section, a duty of 0 kW is observed at interval 4. When no heat is transferred between a hot stream at a temperature above the pinch to a cold stream at a temperature below the pinch, the pinch point is determined. Therefore, at interval 4, the pinch temperature of 130°C is observed.

1. Minimum temperature approach

DTmin20 °C

2. Stream data

Stream No.	Actual temperature (°C)		Interval temperature (°C)		Heat capacity flow rate CP (kW/°C)	Heat load (kW)
	Source	Target	Source	Target		
1	280	60	270	50	30	6600
2	180	20	170	10	40	6400
3	20	160	30	170	50	7000
4	120	260	130	270	55	7700
5	0	0	10	10	0	0
6	0	0	10	10	0	0
7	0	0	10	10	0	0
8	0	0	10	10	0	0

3. Problem table

Interval	Interval temp (°C)	Interval DT (°C)	Sum CPc - sum CPh (kW/°C)	dH (kW)	Cascade (kW)	(kW)
	270				0	3900
1	270	0	0	0	0	3900
2	170	100	25	2500	-2500	1400
3	170	0	25	0	-2500	1400
4	130	40	35	1400	-3900	0
5	50	80	-20	-1600	-2300	1600
6	30	20	10	200	-2500	1400
7	10	20	-40	-800	-1700	2200
8						

Figure 7: Summary of pinch analysis. Pinch temp is 130°C (interval 4).

8. Heat Exchanger Network Design

To design an optimized process for maximum heat recovery, a heat exchanger network design is developed. From the stream data table in Figure 7, a grid of the two hot streams and two cold streams and their corresponding source and target temperatures and heat loads is created. Then the streams are matched one hot stream to one cold stream according to heat capacity values. Above the pinch, the Cp value of the cold stream must be higher than the Cp value of the hot stream. Below the pinch, the Cp value of the hot stream must be higher than that of the cold stream. The streams are matched with heat exchangers to create efficiency in heating. As the hot stream passes through the exchanger, it heats up the cold stream to the target temperature, and simultaneously the target temperature of the hot stream is also reached.

Figure 8 represents the final network design of the process. Pinch temperatures of 140°C and 120°C are used as the new target and source temperatures for each side of the pinch. The following calculations were used to determine the enthalpy changes for above the pinch:

$$\Delta H = mC_p\Delta T$$

$$\text{Hot 1: } \Delta H = \left(30 \frac{\text{kW}}{^\circ\text{C}}\right)(280 - 140)^\circ\text{C} = 4200 \text{ kW}$$

$$\text{Hot 2: } \Delta H = \left(40 \frac{\text{kW}}{^\circ\text{C}}\right)(180 - 140)^\circ\text{C} = 1600 \text{ kW}$$

$$\text{Cold 1: } \Delta H = \left(50 \frac{\text{kW}}{^\circ\text{C}}\right)(160 - 120)^\circ\text{C} = 2000 \text{ kW}$$

$$\text{Cold 2: } \Delta H = \left(55 \frac{\text{kW}}{^\circ\text{C}}\right)(260 - 120)^\circ\text{C} = 7700 \text{ kW}$$

The same method was used to calculate the enthalpies below the pinch:

$$\text{Hot 1: } \Delta H = \left(30 \frac{\text{kW}}{^\circ\text{C}}\right)(140 - 60)^\circ\text{C} = 2400 \text{ kW}$$

$$\text{Hot 2: } \Delta H = \left(40 \frac{\text{kW}}{^\circ\text{C}}\right)(140 - 20)^\circ\text{C} = 4800 \text{ kW}$$

$$\text{Cold 1: } \Delta H = \left(50 \frac{\text{kW}}{^\circ\text{C}}\right)(120 - 20)^\circ\text{C} = 5000 \text{ kW}$$

Above the pinch, Hot 1 and Cold 2 are matched and Hot 2 and Cold 1 are matched. The exchanger on Hot 1 provides 4200 kW to Cold 2 and Hot 2 provides 1600 kW to Cold 1. However, Cold 2 requires 7700 kW to heat the stream to from 120°C to 260°C and Cold 1 requires 2000 kW to heat the stream from 120°C to 160°C. Therefore, a 3500-kW heater is inserted in Hot 2 and a 400-kW heater is inserted in Hot 1.

Below the pinch, Cold 1 was separated into two streams a and b in order to allow stream matching. The C_p values of Hot 1, 30 kW/°C, and Hot 2, 40 kW/°C, are not greater than the C_p values of 50 kW/°C for Cold 1. Recalculating the heat capacity determines a new heat capacity 26 kW/°C for Cold 1a and 24 kW/°C for Cold 1b. Cold 2 was neglected because 120°C is already in the pinch. Hot 1 and Cold 1b are matched and Hot 2 and Cold 1a are matched. The heat exchanger on Cold 1a only gives 2600 kW to Hot 2, which requires 4800 kW to cool the stream from 140°C to 20°C. Thus, a cooler providing 2200 kW is inserted in Hot 2. Hot 1 and Cold 1b provide equal exchange of 2400 kW.

$$\text{Cold 1a: } \Delta H = \left(26 \frac{\text{kW}}{^{\circ}\text{C}}\right) (120 - 20)^{\circ}\text{C} = 2600 \text{ kW}$$

$$\text{Cold 1b: } \Delta H = \left(24 \frac{\text{kW}}{^{\circ}\text{C}}\right) (120 - 20)^{\circ}\text{C} = 2400 \text{ kW}$$

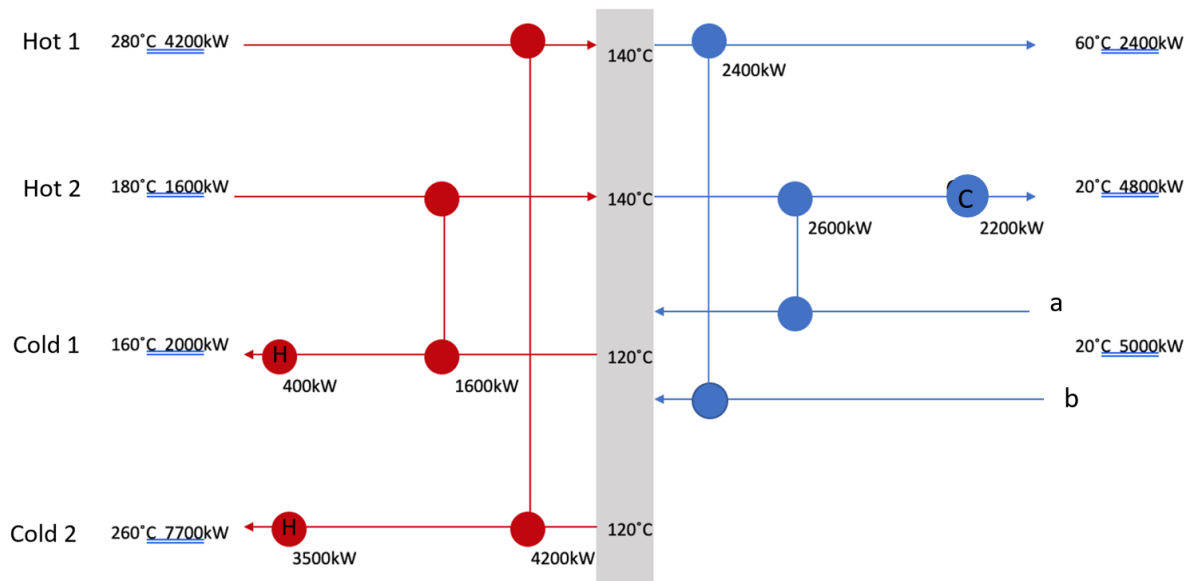


Figure 8: Matching of Hot and Cold Streams

9. Process Flow Diagram (PFD) – After Integration

After heat integration, an optimized process with four heat exchangers, two heaters, and one cooler was designed. The PFD in Figure 9 shows the arrangement of heat exchangers and utilities in the process. Stream 3 is split into two streams to allow the exchange of heat between the cold streams and the hot streams from the distillation column. Cold streams 3a and 3b are heated to 120°C and hot streams 12 and 13 are cooled to 75°C and 60°C, respectively. Streams 4 and 11 exchange heat to generate a temperature of 152°C in stream 5 and 140°C in stream 13. Utility 1 heats the stream going into the reactor to 160°C. The reactor outputs a stream temperature of 120°C which is then heated to 196°C by utility 2. In turn, stream 10 is cooled from 180°C to 140°C. Utility 3 completes the final cooling of the distillate product to 20°C [2]. A summary of the stream data is found in Table 4.



Stream	T (°C)	Flow (kg/s)	Cp (kJ/kg °C)	Utility	Δ H(kW)	Utility Type
1	20	5	5	-	-	-
2	20	5	5	-	-	-
3	20	10	5	-	-	-
3a	20	5.2	5	-	-	-
3b	20	4.8	5	-	-	-
4a	120	5.2	5	-	-	-
4b	120	4.8	5	-	-	-
4	120	10	5	-	-	-
5	152	10	5	1	400	MP steam
6	160	10	5			
7	120	10	5.5	-	-	-
8	196	10	5.5	2	3500	MP steam
9	260	10	5.5			

10	180	4	10	-	-	-
11	280	6	5	-	-	-
12	140	4	10	-	-	-
13	140	6	5	-	-	-
14	75	4	10	3	2200	Chilled Water
15	60	6	5	-	-	-
16	20	4	10	3	2200	Chilled Water

10. Utility Cost

When selecting utilities, the target stream temperatures and cost of the utility are considered. To ensure the ability of the utility to heat the stream to the target temperature, an allowable temperature difference of 10°C was assumed [2]. Before heat integration, medium pressure (MP) steam, cooling water, and chilled water are utilized to heat the system. MP steam at 300°C has a max allowable heating of 290°C and is capable of heating both cold streams to 160°C and 260°C. Cooling water is limited to cooling to temperatures above 40°C and is used to cool hot stream 1 to 60°C. Chilled water is capable of cooling to temperatures above 5°C and is to cool hot stream 2 to 20°C. After heat integration, MP steam and chilled water are used to reach the target temperatures of 160°C for utility 1, 260°C for utility 2, and 20°C for utility 3 [5]. Summaries of the chosen utilities are provided in Table 5 and Table 6.

Cost of utilities for both before and after heat integration were calculated using the annual prices per kilowatt (kW) from *Chemical Engineering Design* by Towler and Sinnott and *Utility Systems* PowerPoint slides by Betul Bilgin [1][5]. Medium pressure (MP) steam annual cost per kW was calculated to be \$114.832 using the following equations:

$$1 \text{ kW} = \left(1 \frac{\text{kJ}}{\text{s}}\right) \left(\frac{3600 \text{ s}}{1 \text{ hr}}\right) \left(\frac{8000 \text{ hrs}}{1 \text{ yr}}\right) \left(\frac{1}{3025 \frac{\text{kJ}}{\text{kg}}}\right) = 9520.66 \text{ kg/yr}$$

$$\text{At } 300^\circ\text{C } h_g = 3025 \text{ kJ/kg}$$

$$9520.66 \frac{\text{kg}}{\text{yr}} \left(\frac{2.205 \text{ lb}}{1 \text{ kg}}\right) \left(\frac{\$5.47}{1000 \text{ lb}}\right) = \$114.832/\text{yr}$$

The annual costs per kW for cooling water and chilled water are \$18.10 and \$129.60, respectively [1]. Using the annual cost per kW and duty, the following before integration costs were calculated:

$$\text{Utility 1: } \frac{\$114.832}{(\text{kW})\text{yr}} (7000 \text{ kW}) = \$803,824.00/\text{yr}$$

$$\text{Utility 2: } \frac{\$114.832}{(\text{kW})\text{yr}} (7700 \text{ kW}) = \$884,207.00/\text{yr}$$

$$\text{Utility 3: } \frac{\$129.60}{(\text{kW})\text{yr}} (6400 \text{ kW}) = \$829,440.00/\text{yr}$$

$$\text{Utility 4: } \frac{\$18.10}{(\text{kW})\text{yr}} (6600 \text{ kW}) = \$119,460.00/\text{yr}$$

Table #5: Utility Costs Before Heat Integration.

Utility	Duty (kW)	Cost	Annual Cost per kW	Utility Cost Per Year
1 – MP Steam	7000	\$5.47/1000 lb	\$ 114.83	\$ 803,824.00
2 – MP Steam	7700	\$5.47/1000 lb	\$ 114.83	\$ 884,207.00
3 – Chilled Water	6400	\$4.50/GJ	\$ 129.60	\$ 829,440.00
4 – Cooling Water	6600	\$0.10/1000 gal	\$ 18.10	\$ 119,460.00
			TOTAL	\$2,636,930

After heat integration, the costs of the three utilities were calculated using the same method as above:

$$\text{Utility 1: } \frac{\$114.832}{(\text{kW})\text{yr}} (400 \text{ kW}) = \$45,932.00/\text{yr}$$

$$\text{Utility 2: } \frac{\$114.832}{(\text{kW})\text{yr}} (3500 \text{ kW}) = \$401,912.00/\text{yr}$$

$$\text{Utility 3: } \frac{\$129.60}{(\text{kW})\text{yr}} (2200 \text{ kW}) = \$285,120/\text{yr}$$

Table #6: Utility Costs After Heat Integration.

Utility	Duty (kW)	Cost	Annual Cost per kW	Utility Cost Per Year
1 – MP Steam	400	\$5.47/1000 lb	\$ 114.83	\$ 45,932.80
2 – MP Steam	3500	\$5.47/1000 lb	\$ 114.83	\$ 401,912.00
3 – Chilled Water	2200	\$4.50/GJ	\$ 129.60	\$ 285,120.00
			TOTAL	\$732,965

Compared to the utility cost before integration, the after-integration utility cost is 28% of the original cost and saves \$1,903,970 per year. Using three utilities and four heat exchangers eliminates the cost of the cooling water and decreases the duties of the utilities. MP steam is used to meet the heat load of 3900 kW whereas in the before process, a total of 14,700 kW was required. This reduces the total cost by almost half of the original cost. Chilled water is needed to provide a heat load of 2200 kW instead of 6400 kW, further reducing the total cost by over \$500,000 per year.

11. Project Charter – Part 2

Title	Goal	In Scope	Out of Scope
Production of MTBE	Explain the process of production of MTBE	Model and Simulate the production of MTBE with ASPEN	Safety Hazard concerns for MTBE production to the environment
Deliverables		Team Members	
- Design Plan - BFD - Feeds and Mass balances for each stage - PFD - ASPEN Simulation		- Huda Usman - Milos Markovic - Kellen Krupa - Joel Finlon	

12. Design Plan – Part 2

Task ID	Task	Start Date	Finish Date	Member assigned
1	Task Delegation	10/5	10/6	All
2	Start research	10/5	10/8	Huda & Milos
3	BFD	10/6	10/7	Kellen
4	PFD	10/6	10/7	Joel
5	Material Balance	10/7	10/8	Joel and Milos
6	Simulation of shortcut	10/8	10/11	Huda and Kellen
7	Simulation of Rigorous	10/9	10/11	Milos and Huda
8	Varying parameter explanations	10/11	10/11	Joel and Huda
9	Unification/Proofreading	10/12	10/12	All

13. Introduction - Methyl Tert Butyl Ether: The Distillation Column Retriever

In this project we are designing the process to produce 7000 kg/hr pure Methyl Tert-Butyl Ether (MTBE). MTBE is used as an additive in gasoline and is generated by the reaction between isobutene with methanol [2]. The process consists of two feed streams one containing pure methanol and the other consists of C4 compounds which includes, n-butane, 1-butene, 2-butene, and isobutene. Both streams in our design are mixed before going to the reactor. Inside the reactor methanol and isobutene react together to give MTBE and the unreacted C4 compounds comes out of the reactor. Then the reactor outlet stream is introduced into the 1st distillation column where mostly pure MTBE is exited out from the bottom. Methanol and C4 compounds are exited from the top of the distillation column 1. The top stream from the distillation column 1 is introduced into the absorber to separate methanol and C4 compounds using water solvent. C4 compounds exit out from the top of the absorber and can be used for other applications such as fuel gas [2]. 99 % methanol exits out from the bottom of the absorber and is then introduced into the 2nd distillation column where it is separated from water and is then recycled into the mixer. 10 % purge water is exited out from the bottom of the 2nd distillation column whereas remaining water is recycled into the absorber to reduce impurities [2].

14. Unit Balances

14.1 Balance around the mixer

Balances around the mixer include the fresh methanol and the fresh C4 compounds as well as a methanol recycle stream from the very last column in the process. Such materials need to be mixed before entering the reactor and this preliminary step allows for a constant specified flow for the methanol into the reactor as that will not change regardless of the flow from the recycle stream.

Table #7: Flow Rates for Mixer

S2, C4 Compounds	Mole Fractions	Flow Rate (kmol/hr)	Flow Rate (kg/hr)
Iso-butene	0.49	82.28	4616
n-butane	0.02	3.358	195.2
1-butene	0.31	52.05	2921
2-butene	0.18	30.22	1696
S1, Methanol	1.0	79.92	2900
Methanol Recycle	1.0	10.59	339.3

14.2 Balance around the reactor

At this stage in the process, the now mixed materials enter the reactor and produce MTBE. Excess methanol was used to suppress side reactions. Only the mole flows for the isobutene and methanol will change leaving the reactor, while the other C4 compounds (nButane, and 1Butene, 2Butene) will remain unchanged since they are inert for the reaction.

Table #8: Flow Rates for Reactor

Reactor	Flow Rate IN (kmol/hr)	Flow Rate OUT (kmol/hr)	Flow Rate (kg/hr)
Iso-butene	82.28	2.468	138.5
C4 Compounds	85.64	85.64	4812
Methanol	90.50	10.58	342.7
MTBE	0	79.81	7035

14.3 Balance around the Distillation column 1

Around the first distillation column is where our desired product flow for MTBE was specified. Using mass balances and back calculation to find the inflow from the desired specification, the other materials flows were found. Important Assumptions were made with the mass balance calculations such as that the bottoms product was pure MTBE and every other compound left the top of the column. This assumption cannot be made in practicality since Methanol and MTBE form an azeotrope.

Table #9: Flow Rates for Distillation Column 1

Distill column 1	Flow Rate IN (kmol/hr)	Flow Rate TOPs (kmol/hr)	Flow Rate (kg/hr)
Iso-butene	2.468	2.468	138.5
C4 Compounds	85.64	85.64	4812
Methanol	10.69	10.69	342.7
MTBE		0.3990	35.18
MTBE	79.81		7035
		Flow Rate BOTs (kmol/hr)	
		79.41	7000

14.4 Balance around the Absorber

In this section of the process, Methanol and Water are separated from the C4 components and isobutene to be used as recycle streams back into the original process. A freshwater feed was added at this stage of the process for absorption.

Table #10: Flow Rates for Absorber

Absorber	Flow Rate IN (kmol/hr)	Flow Rate TOPs (kmol/hr)	Flow Rate (kg/hr)
Iso-butene	2.468	2.468	138.5
C4 Compounds	85.64	85.64	4812
MTBE	0.3990	0.3990	35.18
Methanol		0.1069	3.427
Methanol	10.69		342.7
		Flow Rate BOTs (kmol/hr)	
Methanol		10.58	339.3
Water		60	1081

14.5 Balance around the Distillation Column 2

This last distillation column serves as a Methanol and Water recovery. This distills Methanol from water and send it to the beginning of the process to add to the total methanol inflow to the mixer. The water was then sent back to the absorption column and added with the freshwater feed for carbon absorption.

Table #11: Flow Rates for Distillation Column 2

Distill column 2	Flow Rate TOPs (kmol/hr)	Flow Rate (kg/hr)
Methanol Recycle	10.58	339.3
	Flow Rate BOTs (kmol/hr)	
Water	60.01	1081
Water purge	6.001	108.1
Water Recycle	54.00	972.9

15. Aspen Simulation of MTBE with Shortcut Column

15.1 Feed Specifications

ASPEN Plus is utilized for the simulation of distillation column 1. First, shortcut model was run using the column DISTWU. The method used to run the simulation was NRTL because of the presence of polarity and hydrogen bonding in interested compounds. The feed specifications used for the shortcut model is shown below.

The screenshot shows the 'Specifications' tab in the ASPEN Plus interface. The 'Flash Type' is set to 'Pressure' and 'Vapor Fraction'. The 'State variables' section includes: Temperature (103 C), Pressure (27 bar), Vapor fraction (0), Total flow basis (Mole), Total flow rate (178.614 kmol/hr), and Solvent (empty). The 'Composition' section shows a 'Mole-Flow' table with the following data:

Component	Value
METHANOL	10.7
N-BUTANE	3.36
1-BUT-01	52.0528
2-BUT-02	30.224
ISOBUT-01	2.468
MTBE	79.809
Total:	178.614

Figure 7: ASPEN Plus Feed Input

15.2 Column Specifications

The feed was given in a saturated liquid phase therefore the vapor fraction was given to be equal to zero. Pressure 27 bar was chosen because it is less than the critical pressure of MTBE [6] so that we can have the majority of MTBE coming out from the bottom of the distillation column.

The pressures of the condenser and reboiler was set to 26.46 bar and 27 bar respectively. The input specifications into the DISTWU column is shown below.

Specifications	Calculation Options	Convergence	Information
Column specifications ☐ Number of stages: ☒ Reflux ratio: -1.15			
Pressure Condenser: 26.46 bar Reboiler: 27 bar			
Key component recoveries Light key: Comp: METHANOL Recov: 0.99 Heavy key: Comp: MTBE Recov: 0.0055		Condenser specifications ☒ Total condenser ☐ Partial condenser with all vapor distillate ☐ Partial condenser with vapor and liquid distillate Distillate vapor fraction: 0	

Figure 8: ASPEN Plus DSTWU Column Input

15.3 Stream Results

The stream and block results after running the simulation is shown below. S2 stream shows the distillate leaving the top of the column and S3 shows the bottom product containing pure MTBE.

		Units	S1	S2	S3
▶	N-BUTANE	KMOL/HR	3.36	3.3593	0.000707037
▶	1-BUT-01	KMOL/HR	52.0529	52.0462	0.00665333
▶	2-BUT-02	KMOL/HR	30.224	30.1972	0.0268801
▶	ISOBUT-01	KMOL/HR	2.468	2.46796	4.23664e-05
▶	MTBE	KMOL/HR	79.8091	0.43895	79.3701
▶	Mole Flow	KMOL/HR	178.614	99.1026	79.5114
▶	Mass Flow	KG/HR	12328.1	5326.31	7001.81
▶	Volume Flow	L/MIN	435.324	204.508	281.388
▶	Temperature	C	154.105	129.699	209.542
▶	Pressure	BAR	27	26.46	27
▶	Vapor Fraction		0	0	0
▶	Liquid Fraction		1	1	1
▶	Average Molecular Weight		69.021	53.7454	88.0604

Figure 9: ASPEN Plus Product Stream Results

15.4 Aspen Simulation Flowsheet

As it can be seen from the above results that the simulation resulted in 99.5 % recovery of MTBE at the bottom. The ASPEN flowsheet of the designed DISTWU column is shown below.

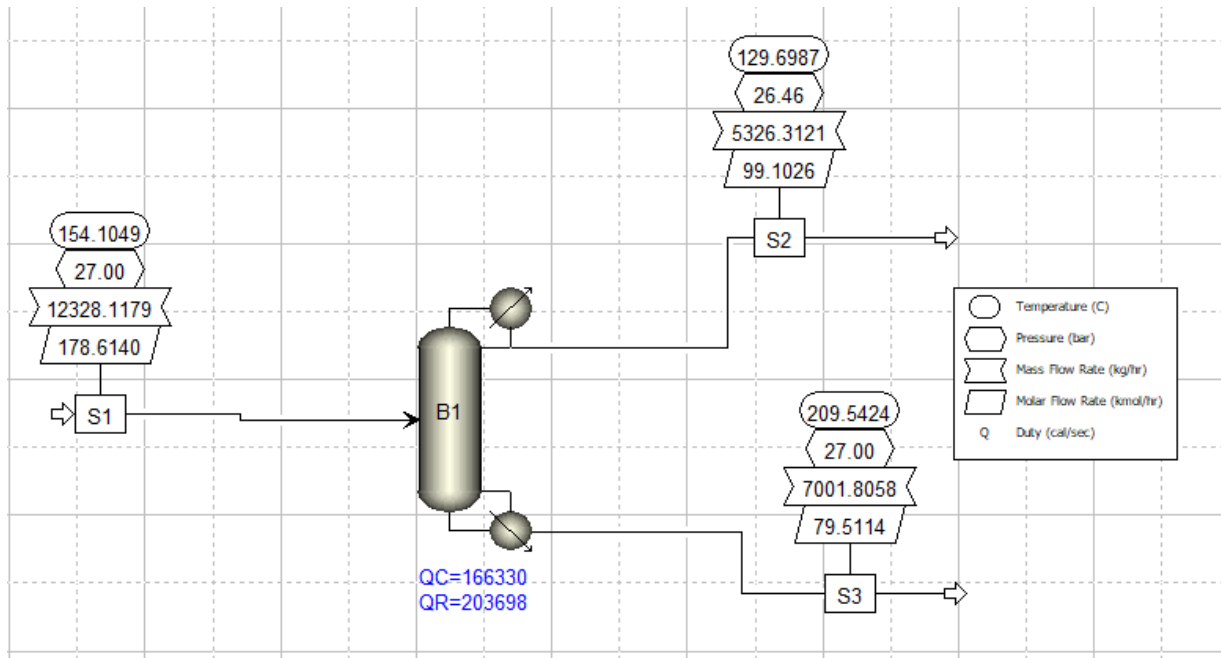


Figure 10: ASPEN Plus Flowsheet for Column Simulation

16. Aspen Simulation of MTBE with Rigorous Column

Results obtained from DISTWU shortcut model is then utilized to run the rigorous model using RadFrac column. The parameters collected are feed stage, reflux ratio, actual number of stages and distillate to feed ratio from the shortcut model and input to the rigorous model. The simulation was run using the following Block Results.

Table #12: DSTWU Obtained Parameters

	Value
Reflux Ratio	0.931185575
Actual number of stages	27.2791904
Feed Stage	18.5756352
Distillate to feed ratio	0.554842161

16.1 Feed Specifications

To run the simulation the condenser pressure was specified to be 27 bar and the feed stage was specified to be 19 (above stage). The following RadFrac specifications were given using the information obtained by shortcut model.

The screenshot shows the ASPEN Plus RadFrac Column Input dialog box. The 'Configuration' tab is active. The 'Setup options' section includes: Calculation type: Equilibrium; Number of stages: 27; Condenser: Partial-Vapor; Reboiler: Kettle; Valid phases: Vapor-Liquid; Convergence: Standard. The 'Operating specifications' section includes: Reflux ratio: 0.931186 (Mole); Distillate to feed ratio: 0.554842 (Mole); Free water reflux ratio: 0. A 'Feed Basis' button is located at the bottom right of the operating specifications section.

Section	Parameter	Value	Unit
Setup options	Calculation type	Equilibrium	
	Number of stages	27	
	Condenser	Partial-Vapor	
	Reboiler	Kettle	
	Valid phases	Vapor-Liquid	
	Convergence	Standard	
Operating specifications	Reflux ratio	0.931186	Mole
	Distillate to feed ratio	0.554842	Mole
	Free water reflux ratio	0	
	Feed Basis		

Figure 11: ASPEN Plus RadFrac Column Input

16.2 Stream Results

The stream results after plugging in these values and running the simulation is shown below.

		Units	S1 ▼	S2 ▼	S3 ▼
▶ Phase:			Liquid	Vapor	Liquid
▶ Component Mole Flow					
▶ NBUTE	KMOL/HR		3.35826	3.34695	0.0113152
▶ ISOBUTE	KMOL/HR		2.46832	2.45908	0.00924342
▶ 1BUTE	KMOL/HR		52.0531	51.7047	0.348445
▶ 2BUTE	KMOL/HR		30.2244	29.2854	0.939021
▶ METHANOL	KMOL/HR		10.6961	9.41185	1.28425
▶ MTBE	KMOL/HR		79.8091	2.89205	76.9171
▶ Mole Flow	KMOL/HR		178.609	99.1	79.5093
▶ Mass Flow	KG/HR		12332.9	5438.13	6894.8
▶ Volume Flow	L/MIN		436.206	2102.57	267.022
▶ Temperature	C		154.141	140.244	204.386
▶ Pressure	BAR		27	27	27
▶ Vapor Fraction			0	1	0
▶ Liquid Fraction			1	0	1
▶ Average Molecular Weight			69.0497	54.8752	86.7168

Figure 12: ASPEN Plus Product Stream Results

From the rigorous model, the % recovery of MTBE was found to be 96.4 % which is less than given in the problem statement. This may be due to the fact that methanol and MTBE is not completely separated in 1st distillation column and there are still some traces of methanol in the stream coming out of the column that is affecting the purity of MTBE.

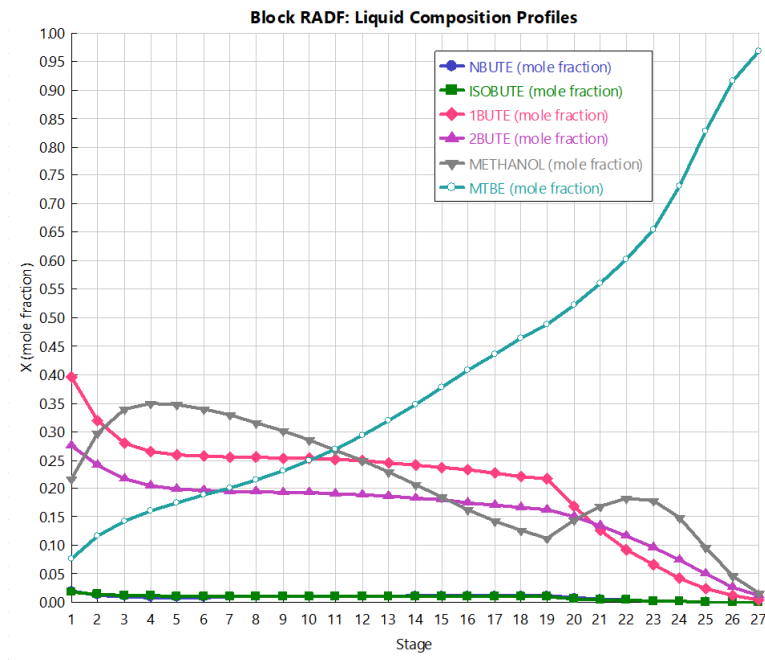


Figure 13: RadFrac Stage Column Results

Size of the Column:

The length of the column was obtained using,

$$L = 2 \text{ ft} * 25 \text{ stages} = 50 \text{ ft} + 10 \text{ ft (reboiler and condenser unit)} = 60 \text{ ft}$$

Also, the diameter of the column was noted to be, 0.910096 m, from the rigorous model.

17. Parameter Variation Results - Rigorous Simulation

The effects of changing reflux ratio, number of stages, column pressure, and feed stage was observed on the composition and recovery of MTBE at the bottom and the composition vs stage graphs were plotted.

17.1 Varying Feed Stage

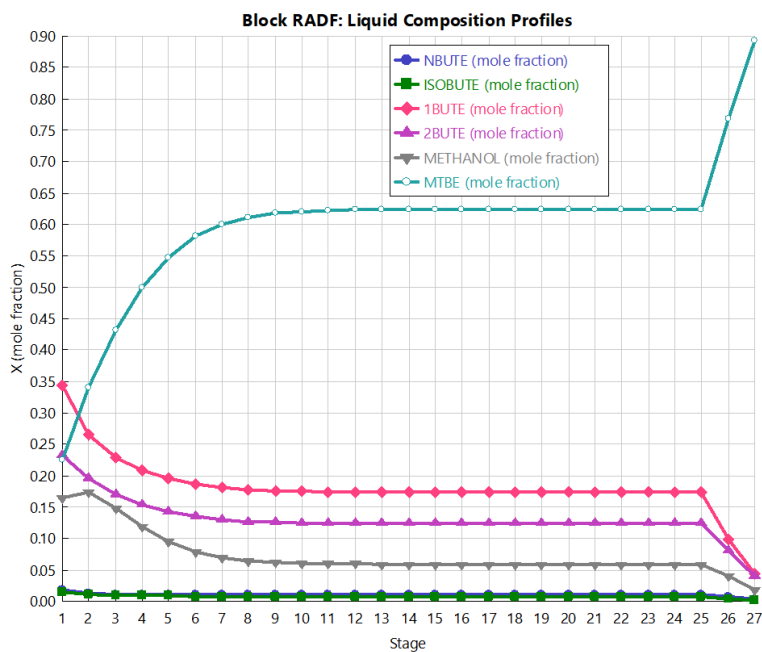


Figure 14: High Feed Stage of 25

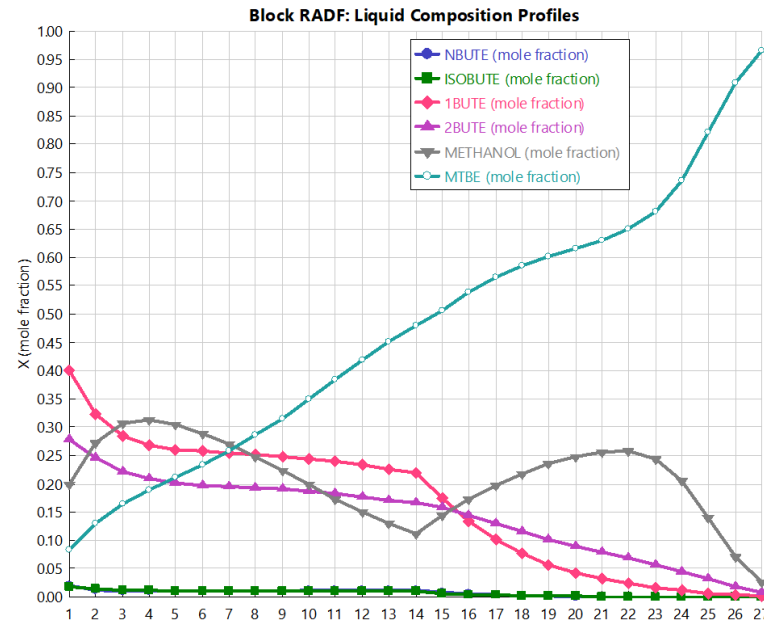


Figure 15: Low Feed Stage of 14

When increasing the feed stage from 19 to 25, the molar composition of the components in the column between stage 4 and 25 do not change. These become wasted trays and if the feed were to be placed so high up relative to the rest of the column then the height of the column could be drastically reduced to accommodate the required stages for molar separation. As observed by the graph, the total purity of the mixture for MTBE decreased as well since MTBE and Methanol did not have enough trays to separate once inputted into the column

When decreasing the feed stage from 19 to 14, the purity of the MTBE in the bottoms of the column decreased. By increasing the feed tray for the stream, MTBE and methanol did not have enough stages to separate fully, but by decreasing the feed stage, the methanol had too many stages and instead of separating further from MTBE, it continued to mix into it because of the two compounds azeotropic properties.

17.2 Varying Column Pressure

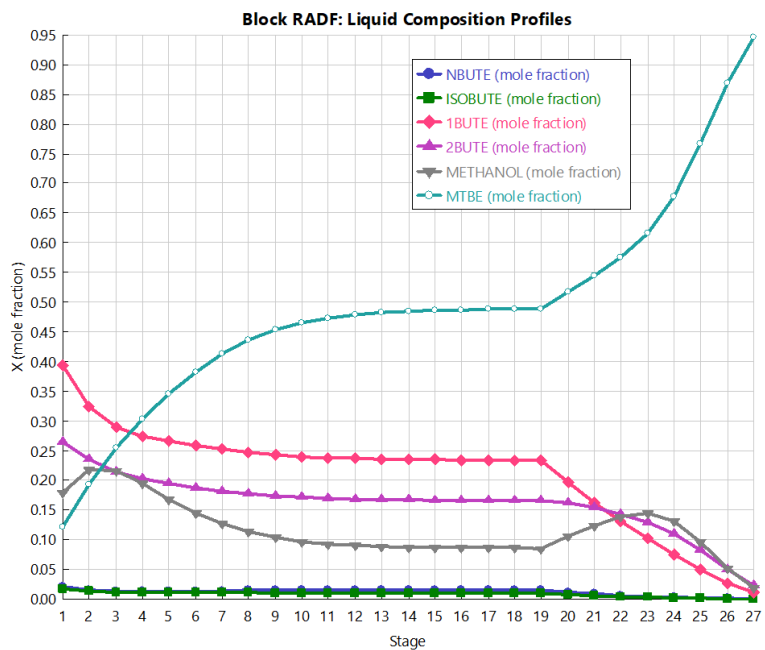


Figure 16: High Column Pressure of 35 bar

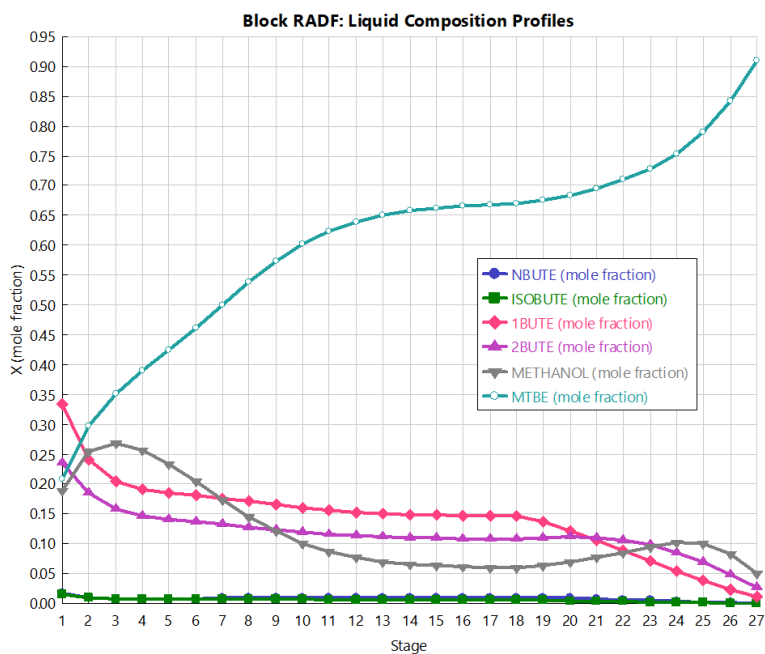


Figure 17: Low Column Pressure of 10 bar

When increasing the column operating pressure from 27 to 35 bar, the tray location for the azeotropic between MTBE and the methanol shifted higher in the column. Instead of being found around tray 12, the increased pressure shifted the azeotrope to tray 2. Along with an increase in column pressure, the trays between stage 11 and 19 became unused and little to no molar composition seemed to change between the components. An increase of pressure further decreased product purity.

When decreasing the column operating pressure from 27 bar to 10 bar, MTBE and methanol azeotrope disappeared. Immediately, the scope of complete product purity for MTBE becomes a viable option since MTBE and methanol can now be separated. While no trays became wasted once the column pressure decreased, the product purity by this action did decrease more so than when the column pressure was increased.

17.3 Varying Reflux Ratio

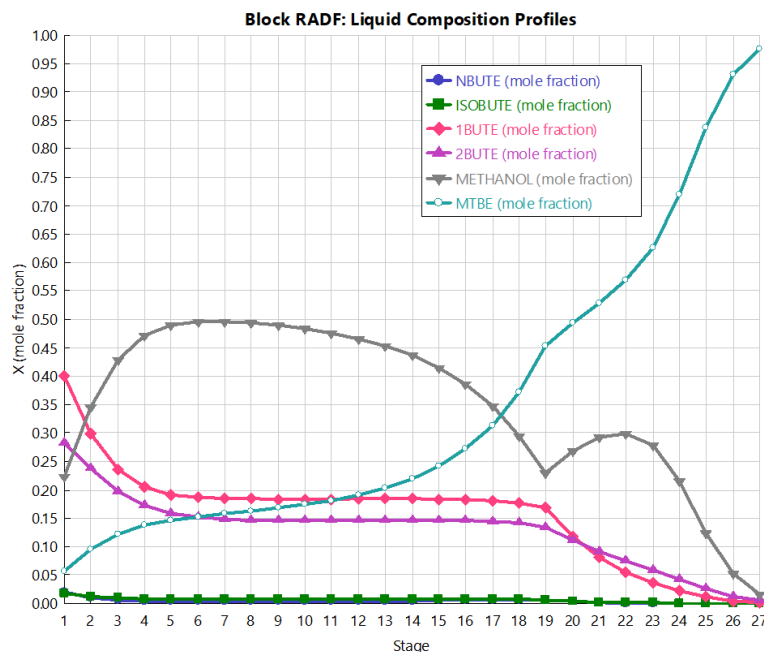


Figure 18: High Reflux Ratio of 2

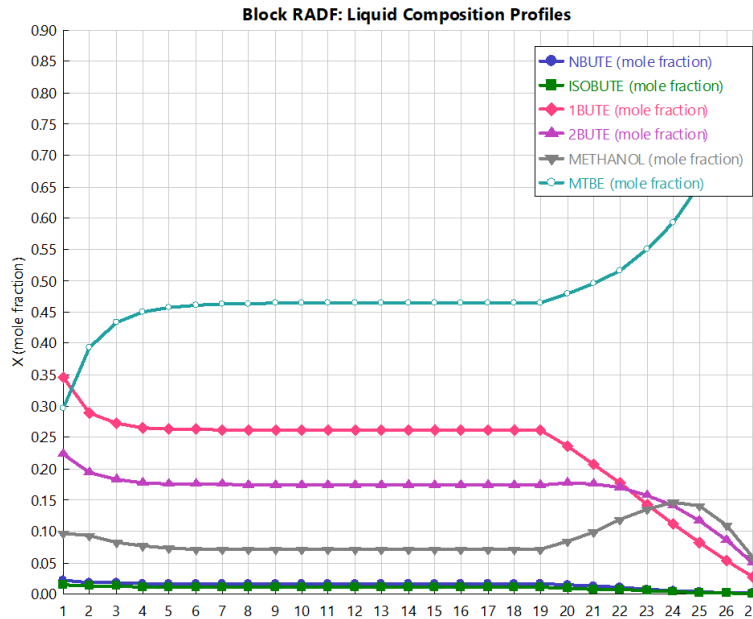


Figure 19: Low Reflux Ratio of 0.4

When increasing the reflux ratio from 0.93 to 2, the final product purity of MTBE in the bottoms remained the same and the azeotrope's tray location didn't move at all. The biggest change from increasing the reflux ratio came about when in the methanol liquid composition throughout the column itself. By decreasing the C4's and MTBE's liquid composition along the column, Methanol's liquid mole fraction almost doubled.

When decreasing the reflux ratio from 0.92 to 0.4, the azeotrope was broken. In this scenario, about 20 stages became wasted and no composition change was observed in the components. The issue by decreasing the reflux ratio is that MTBE's purity in the bottom decreased more than any other process variable change.

17.4 Varying Number of Stages

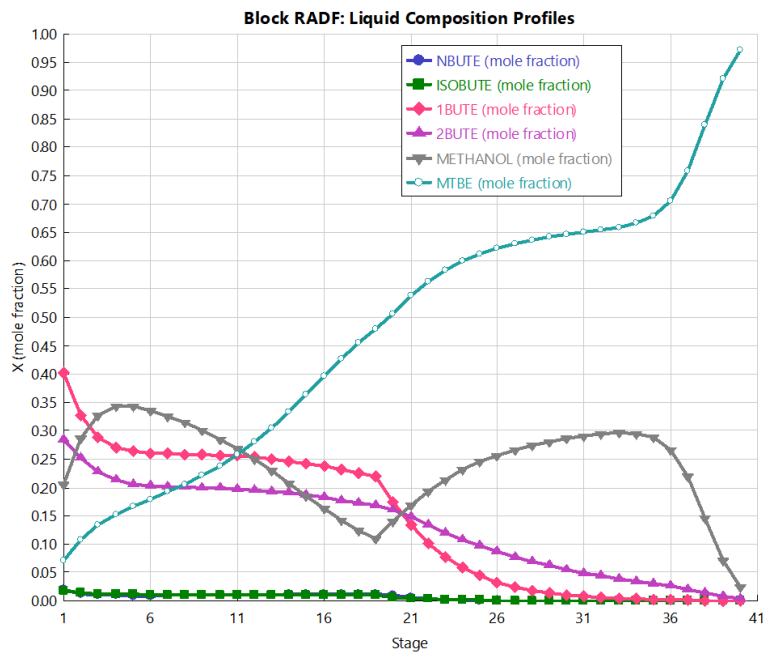


Figure 20: High Total Stage Count of 40

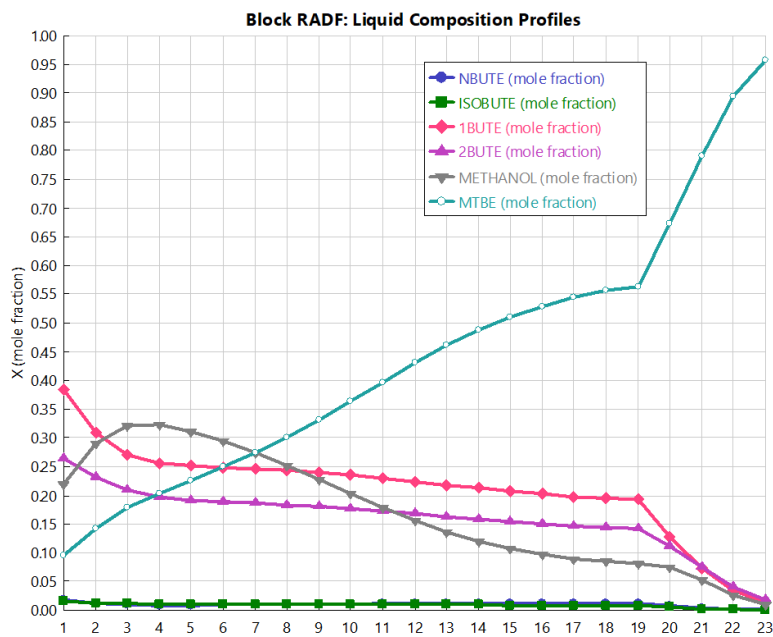


Figure 21: Low Total Stage Count of 23

When increasing the number of stages from 27 to 40, the contact time between the Methanol and MTBE caused more methanol to be mixed into our product. Not only did this result in a higher product impurity, those added higher stages of distillation became wasted as the composition for the C4 compounds and MTBE flattened out right before they needed to leave the column.

When decreasing the number of stages from 27 to 23, product purity decreased further. In this scenario, the tray location for the azeotrope decreased but remained in the same place relative to the entire height of the column. More trays became wasted in this scenario as the most dramatic change in molar composition came about near the very end of the column. With more trays decreased, not enough time was given for the components to separate fully.

18. Azeotrope Search Report Generated by ASPEN

The following report was generated by ASPEN for the presence of an azeotrope after running the rigorous model.

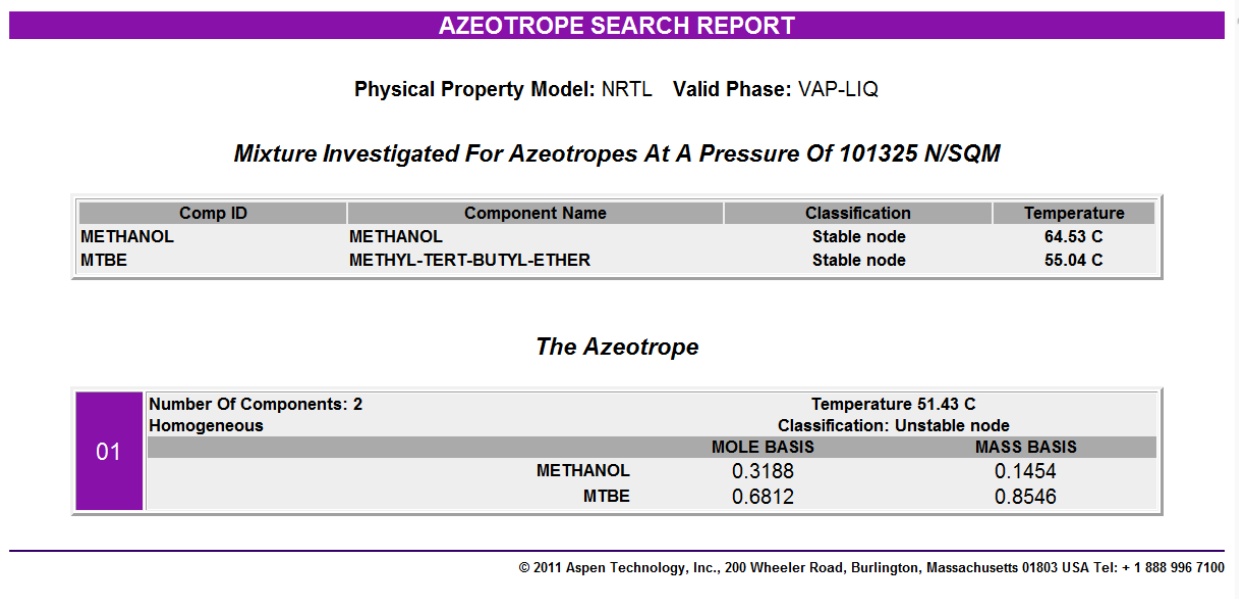


Figure 22: Azeotrope Search Report

Conclusively, our goal was to achieve the given design basis of 7000 kg MTBE / yr in the bottom product stream. However, after running the simulation, we can achieve 6895 kg MTBE / yr in the bottom with the % recovery of 96.4 %. This quantity was achieved using 27 total number of stages and the reflux ratio of 0.93. To enhance the separability of MTBE and methanol, increased reflux ratio and size of column should be utilized to meet the market demand. However, the achieved purity is acceptable since we need improved methods for separating MTBE that can provide around 99.5 % recovery at the bottom such as reactive distillation [7].

19. Block Flow Diagram

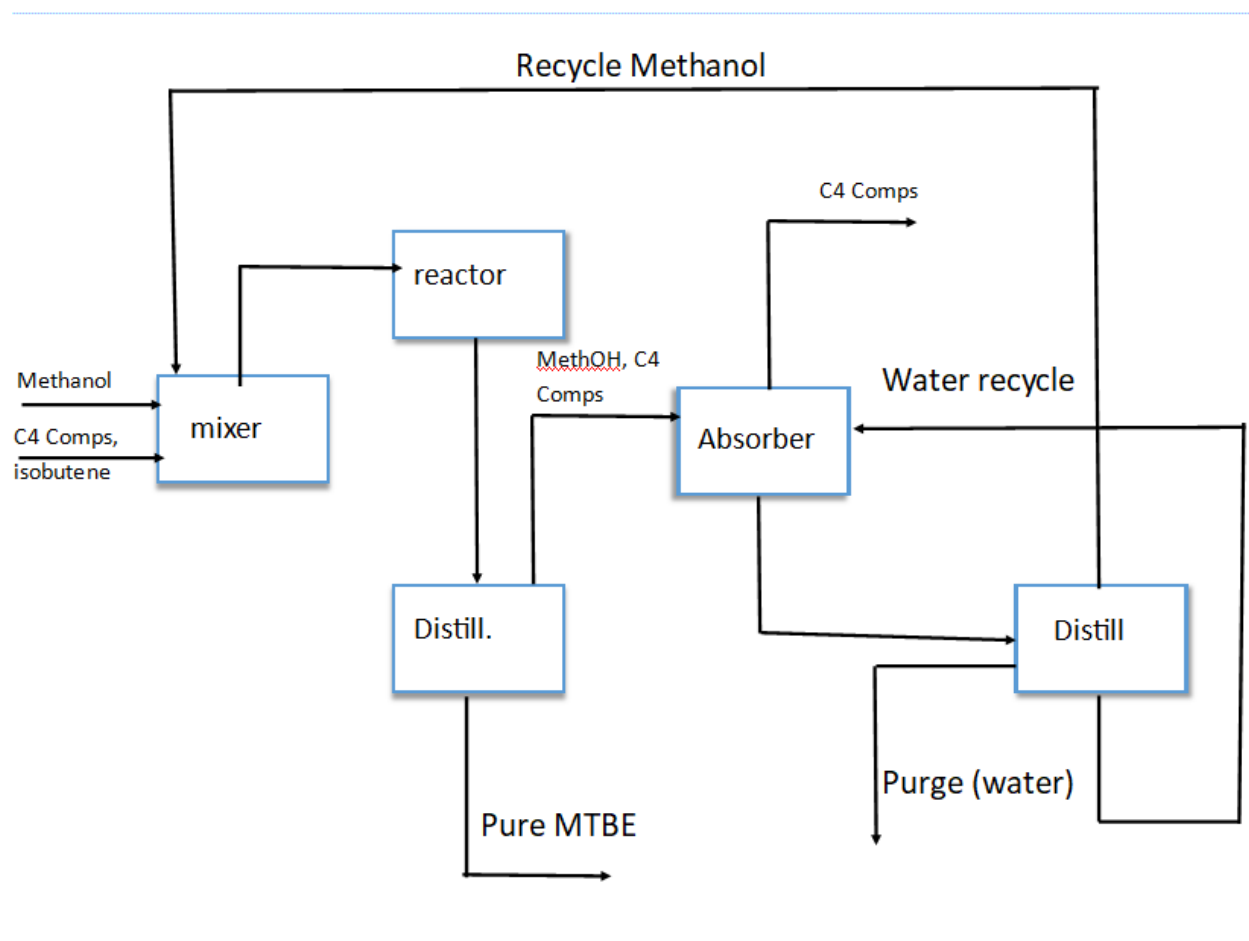


Figure 23: Block Flow Diagram

20. Process Flow Diagram (PFD)

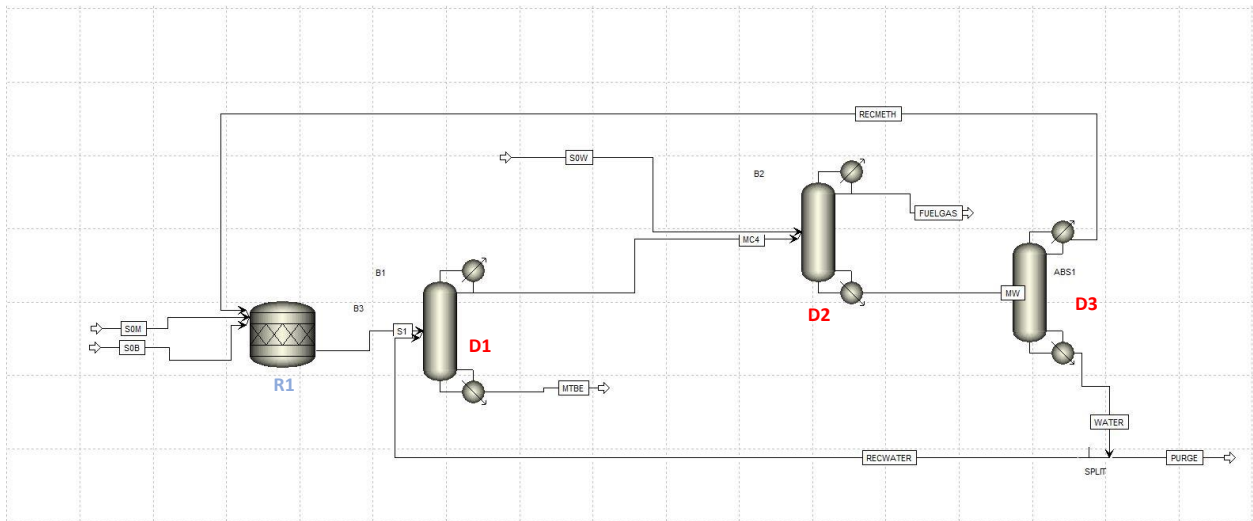


Figure 24: Total Process Flow Diagram

Brief explanation of all equipment in PFD:

1. Reactor – an apparatus or structure that houses a reaction for the conversion of reactants to products
 - a. Reactor 1 (R1): Reactor where MTBE is created via Isobutene and excess Methanol
2. Distillation Column - A tall metal cylinder internally fitted with perforated horizontal plates used to promote phase separation
 - a. Distillation Column 1 (D1): Column where almost Pure MTBE is separated from other feed compounds
 - b. Absorption Column (D2): Absorption Column where the C4 compounds are stripped from Methanol and Water
 - c. Distillation Column 2 (D3): Distillation column separating Methanol and Water for recycle

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